

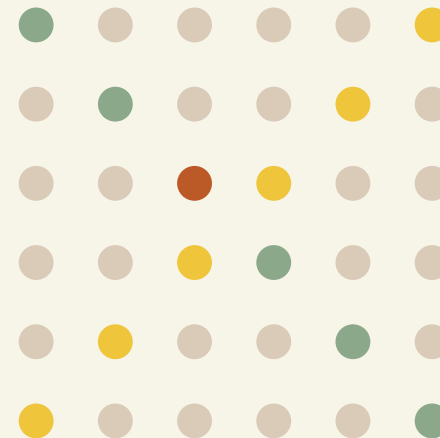
E-WASTE IT WISELY

Mapping Maharashtra's E-Waste Governance Gap

Building a district-level tracking instrument for generation, collection, compliance, and informal-sector integration across the state.

State: Maharashtra **Roll No:** 24102C0086

Live Dashboard: e-waste-dashboard-maharashtra--sidefa3096.replit.app



36 DISTRICTS

The Scale and the Instrument

Why Maharashtra's e-waste stream needs a district-level tracker, not a single state number

THE SCALE

- 134,843 tonnes of e-waste generated statewide each year, modeled from the 2011 Census population base of 112.4 million across 36 districts
- Every resident generates an estimated 1.2 kg of e-waste annually — with a documented upper bound of 2.4 kg/person under the SDG India Index (2021)
- Just three districts — Pune, Mumbai Suburban and Thane — account for roughly a quarter of all e-waste generated in the state
- Generation is split across six equipment categories, from IT & Telecom down to lighting and batteries

THE INSTRUMENT

- An 8-tab, formula-driven tracker anchors every district figure to Census population data and E-Waste (Management) Rules, 2022 targets
- A composite scorecard rolls 7 governance dimensions — generation, collection, recycling, compliance, informal integration, safeguards, outreach — into one score: 82.9 / 100 (“Good”)
- Operational fields without a public benchmark (recyclers, hotspots, inspections) are clearly flagged as illustrative placeholders, not measured data
- Every number recalculates automatically the moment real MPCB registry data replaces a modeled assumption

Maharashtra's e-waste challenge is a data-visibility problem as much as a volume problem — governance requires solving both at once.

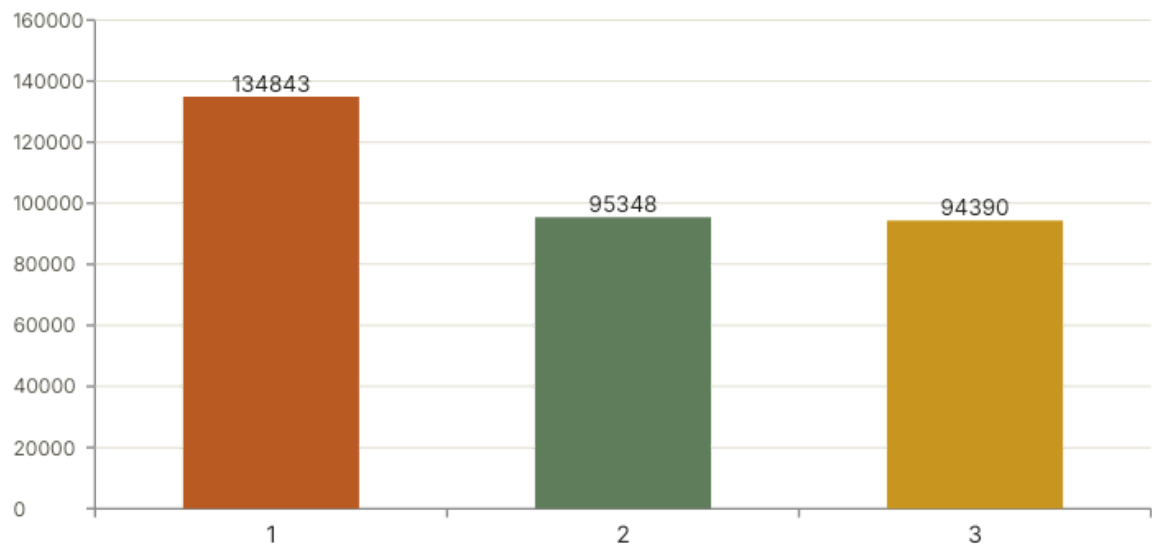
Closing the Collection Gap

Estimated generation vs. modeled collection vs. the current regulatory target

THE NATIONAL CONTEXT

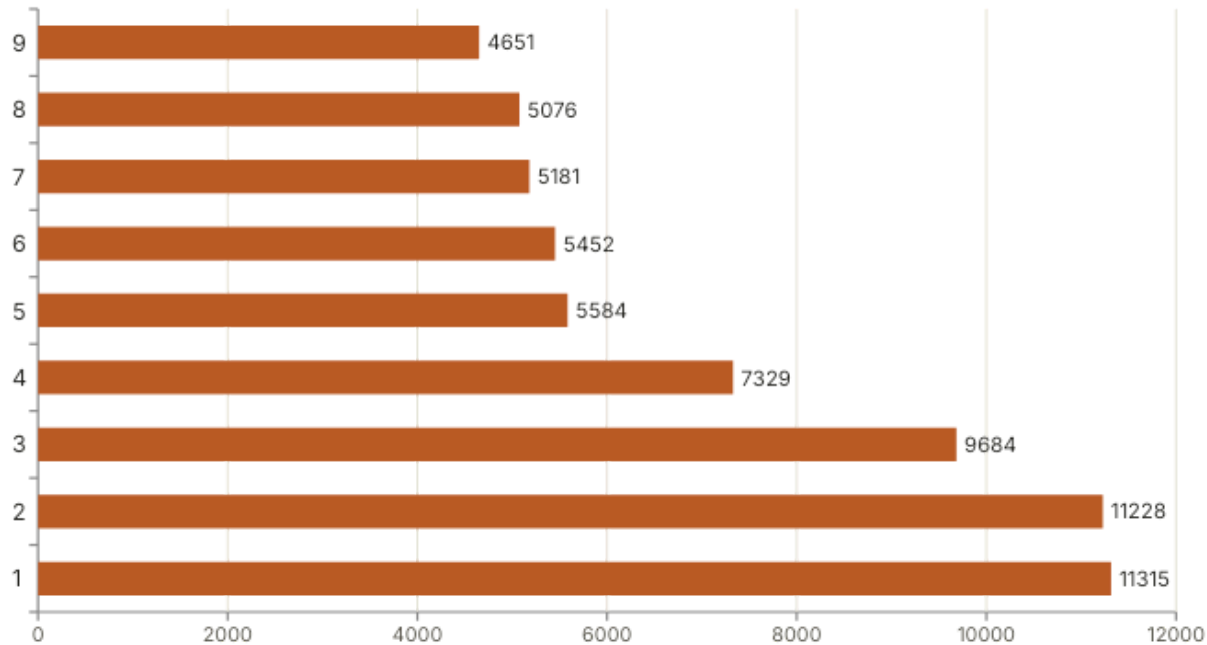
- CPCB recorded ~16,01,155 tonnes of e-waste generated nationally in FY 2021-22
- Formal collection under the EPR portal rose from 61.94% (FY 2023-24) to 70.71% (FY 2024-25)
- Maharashtra's district model applies this 70.71% national rate uniformly — actual district performance will vary with urbanization and enforcement capacity

Generated vs. Collected vs. EPR Target (tonnes/year)



The District Concentration Pipeline

Top 9 of 36 districts by estimated annual e-waste generation

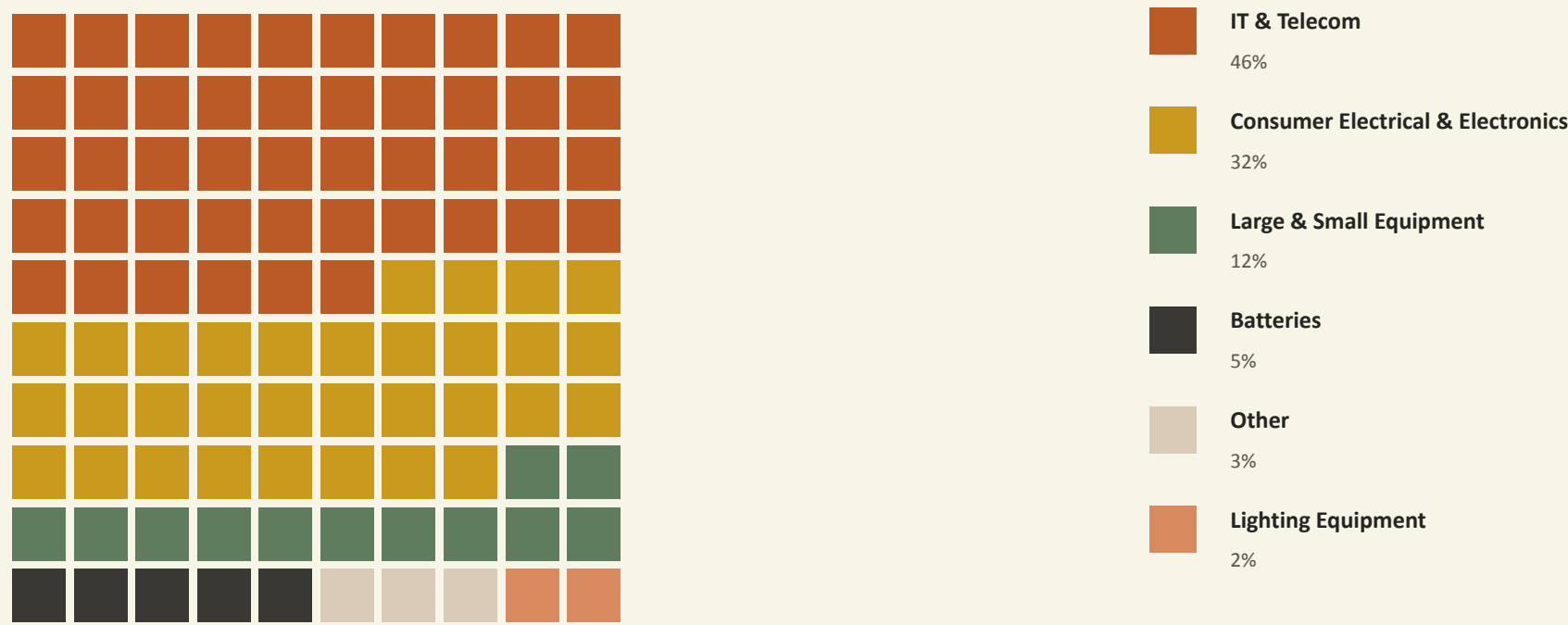


THE INSIGHT

- The Pune – Mumbai Metropolitan Region – Thane belt alone generates over 32,000 t/yr — roughly a quarter of the state total
- Generation tracks population density, not industrial character — the model applies one national per-capita rate to every district
- High-generation districts should be first in line for recycler capacity expansion and EPR-portal audits

The Anatomy of Generation

Statewide e-waste generation, disaggregated across six EEE categories (100 unit grid = 100%)



IT & Telecom and Consumer Electronics together make up 78% of the waste stream — the natural first target for formal collection drives.

Comparison Matrix: Formal vs. Informal Processing

How the registered recycling sector and the informal sector each handle Maharashtra's e-waste

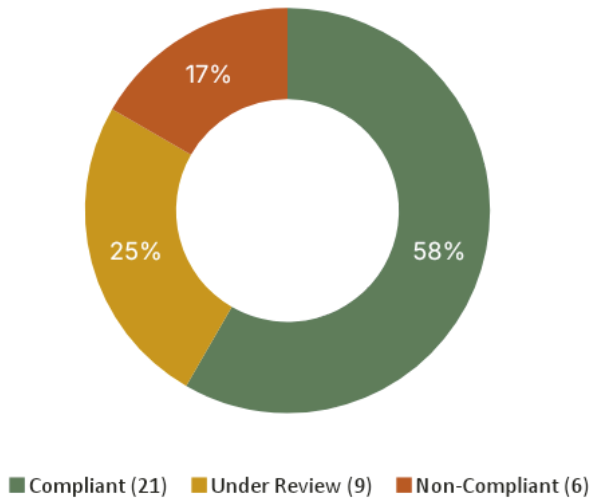
	FORMAL SECTOR	INFORMAL SECTOR
Registered units	36 authorized recyclers / dismantlers (1 per district, illustrative)	Hotspot clusters active in most districts, largely unregistered
Workforce	Employed under CPCB/MPCB authorization	~7,059 estimated informal workers statewide
Formalization	100% operate under valid authorization numbers	Only 2,255 workers (32%) registered or trained
Capacity vs. throughput	38,594 t/yr installed capacity; 24,620 t processed (64% utilized)	Capacity undocumented — no installed-capacity benchmark exists
Outcome	81.8% average material recovery rate	13 of 36 clusters formalized; 12 under monitoring, 6 still active, 5 remediated

The formal sector is capital-efficient at authorized collection, but the informal sector still handles the workforce and reach the formal system hasn't formalized.

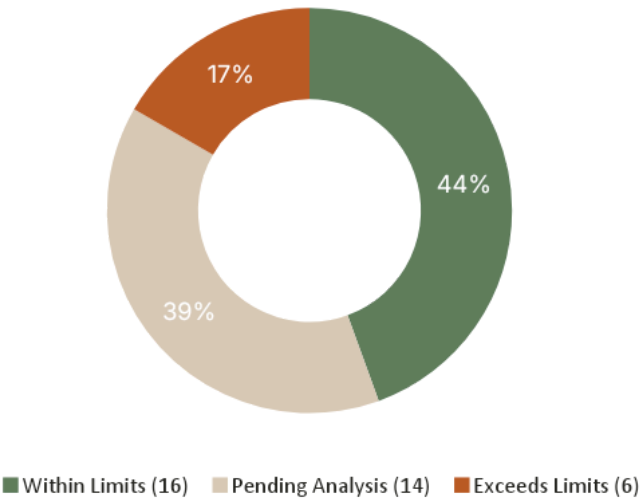
The Enforcement Reality

Producer compliance status and environmental monitoring results across all 36 districts (illustrative field records)

Producer Compliance Status (36 districts)



Environmental / Health Site Monitoring (36 sites)



58% producer compliance and 6 exceedance sites mark the clearest gap between the tracker's structural completeness and verified ground performance.

The Informal Workforce

The human scale behind Maharashtra's informal e-waste recycling clusters

7,059

ESTIMATED INFORMAL WORKERS STATEWIDE

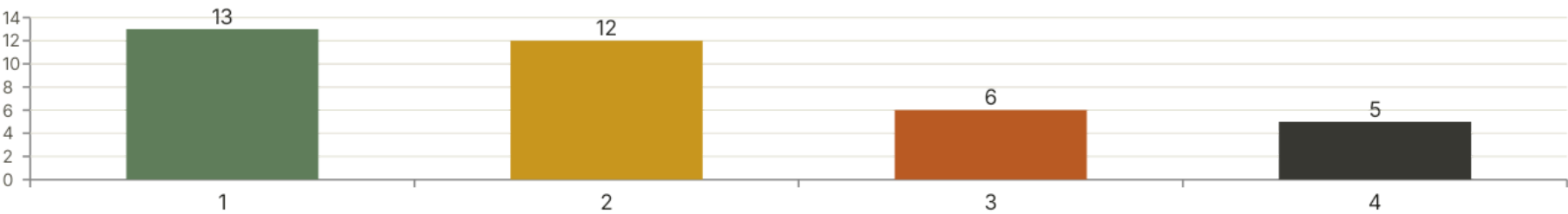
2,255

REGISTERED OR TRAINED (32%)

13 / 36

DISTRICT CLUSTERS FORMALIZED

Informal Cluster Status Across 36 Districts



The Economic and Outreach Engine

Recycling capacity utilization and public awareness reach across the state

RECYCLING CAPACITY		PUBLIC AWARENESS	
Installed capacity	38,594 t / yr	Awareness programs run	36
Quantity processed	24,620 t / yr	Participants reached	30,375
Capacity utilization	63.8%	Health camps conducted	17
Avg. material recovery	81.8%	Beneficiaries screened	2,848
36 authorized recyclers are running at roughly two-thirds of capacity — headroom exists to absorb growth without new licensing.		Every district has run at least one awareness or collection-drive program — the next step is converting reach into registered collection volume.	

The Policy Trajectory vs. Ground Reality

EPR (Management) Rules, 2022 phased targets, national achievement, and the state compliance gap

EPR Collection Target vs. National Achievement

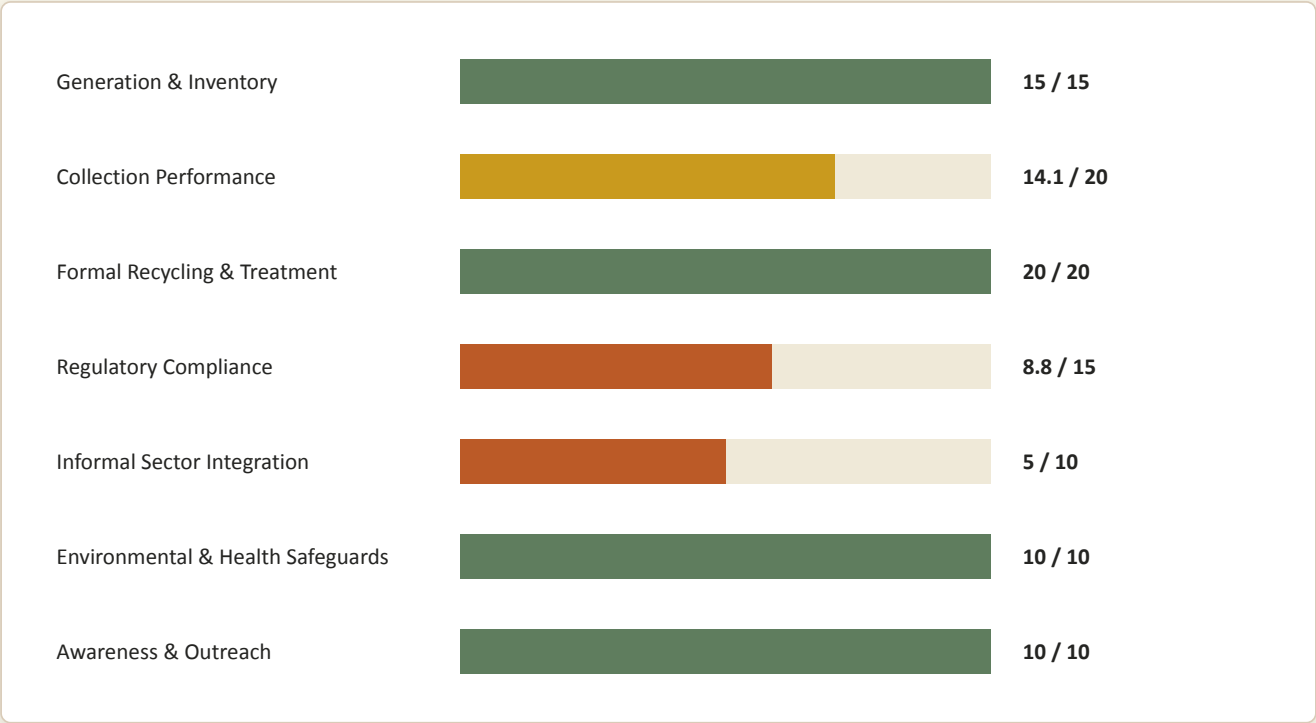


THE DISCONNECT

- National collection is ahead of the regulatory curve — 70.71% achieved against a 70% target
- But Maharashtra's own producer compliance sits at just 58.3%, with 6 of 36 districts flagged non-compliant
- The target rises to 80% from FY 2027-28 — verified district-level tracking has to close this gap before then

The Composite Scorecard: 82.9 / 100

Seven weighted governance dimensions rolled into one baseline score — “Good,” structurally complete



82.9

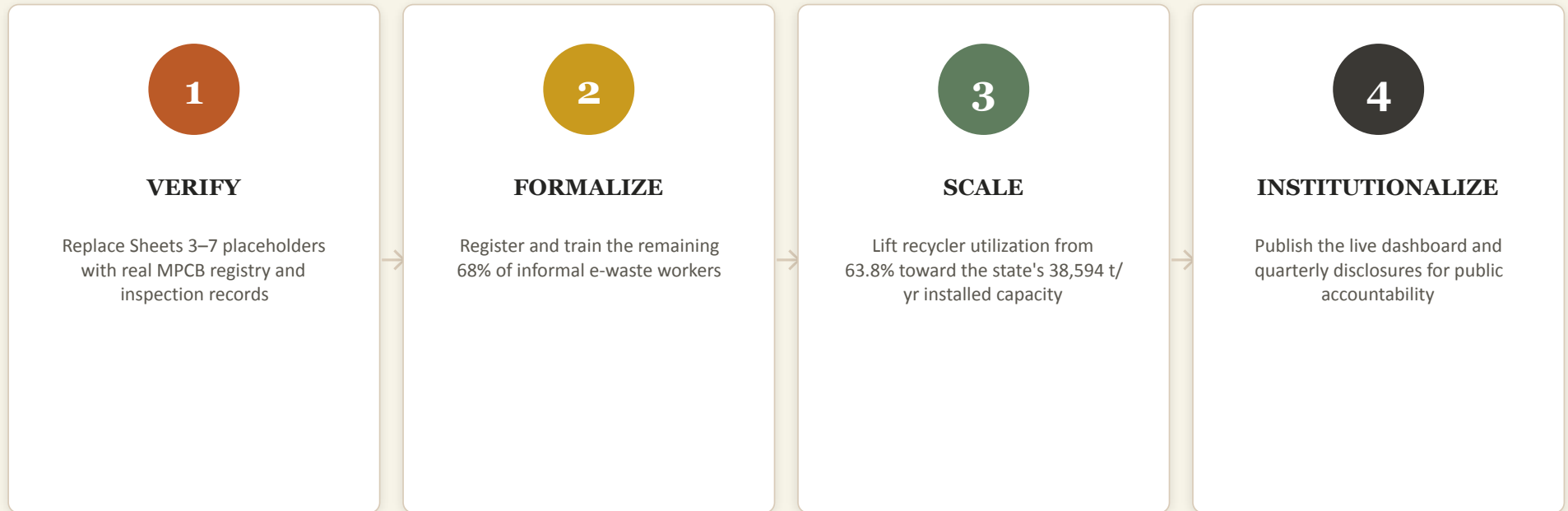
OUT OF 100

GOOD

Reflects a fully populated workbook, including illustrative placeholder data in Sheets 3–7. It measures structural completeness, not verified real-world performance.

Roadmap to a Verified, Field-Ready Tracker

Turning a structurally complete baseline into a governance instrument built on real data



Live dashboard: e-waste-dashboard-maharashtra--sidefa3096.replit.app